



Research Seminar

Group 4

10. Bond Risk Premia with Machine Learning

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April 6, 2020

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Can Machine Learning methods forecast U.S. Treasury bond returns better than traditional linear models?

What is Bond Risk Premium?

The extra return investors earn for holding a long-term bond instead of a short-term one

Traditional approach: Linear models (PCA, PLS, Ridge, Lasso) capture some predictability, but R^2 is modest.

The paper's idea: Non-linear methods - trees, random forests, neural networks - can capture complex interactions between macro variables and bond returns.



Data

- Yield curve: Liu–Wu (2019) zero-coupon Treasury yields.
 - Monthly, 360 maturities (1 month to 30 years), 1961–2025.
- Macro panel: FRED-MD (McCracken–Ng, 2015).
 - 128 monthly U.S. macroeconomic and financial variables.
- Sample period: 1971-08 to 2018-12 (557 monthly observations).
- Target variable: 1-year excess returns for maturities of 2, 3, 4, 5, 7, and 10 years.
- Excess return formula:

$$xr_{t+1}^{(n)} = -(n-1) \left(y_{t+1}^{(n-1)} - y_t^{(n)} \right) + \left(y_t^{(n)} - y_t^{(1)} \right).$$

y - yield at time t, n - remaining maturity

- Forward rates (yield-only inputs): 10 forwards constructed from the yield curve, from 1y to 10y.



Forecasting methods

1. Principal component regressions and partial least squares
2. Penalized regressions: ridge, lasso, elastic net
3. Regression trees
4. Neural networks

Forecasting design and Evaluation Metrics

$$R_{oos}^2 = 1 - \frac{\sum_{t_0=1}^{T-1} \left(x r_{t+1}^{(n)} - \widehat{x r}_{t+1}^{(n)} (\mathcal{M}_s) \right)^2}{\sum_{t_0=1}^{T-1} \left(x r_{t+1}^{(n)} - \overline{x r}_{t+1}^{(n)} \right)^2},$$



1. Principal component regressions and partial least squares

Principal Component Regression (PCR)

- The most common method for forecasting interest rates and bond returns.
- Drawback: when extracting principal components, PCR ignores the target variable, it is purely unsupervised.

Partial Least Squares (PLS)

- An alternative dimension-reduction method.
- Key difference: PLS uses the target when constructing components, it conditions on the joint distribution of predictors and target.
- This makes PLS a supervised compression technique.



2. Penalized regressions: ridge, lasso, elastic net

The problem

When there are many predictors, ordinary least squares overfits the training data and performs poorly out-of-sample.

The idea

Add a penalty to the loss function. This forces the model to give up some in-sample fit in exchange for better generalization.

Method	What the penalty does	Effect on coefficients
Ridge	Penalises large coefficients	Shrinks them towards zero, but never exactly to zero
Lasso	Penalises the absolute size of coefficients	Sets some exactly to zero - variable selection
Elastic Net	Combines both penalties	Balances shrinkage and selection

PCA, PLS, Penalised Linear Regressions, Bond return predictability and the yield curve

What can be seen:

- PCA, PLS miss the signal.
- Ridge, Lasso, Elastic Net outperform PCA and PLS.
- Regularization handles collinearity of forward rates.
- Lasso is the most stable among the three.
- Predictive power grows with maturity.

Models	R^2_{OOS}					
	$xr_{t+1}^{(2)}$	$xr_{t+1}^{(3)}$	$xr_{t+1}^{(4)}$	$xr_{t+1}^{(5)}$	$xr_{t+1}^{(7)}$	$xr_{t+1}^{(10)}$
Panel A: PCA and PLS						
PCA (10 components)	-53.0%	-36.2%	-27.5%	-20.3%	-12.4%	-0.7%
PCA (5 components)	-54.9%	-38.8%	-30.4%	-20.3%	-15.3%	-3.1%
PCA (3 components)	-17.6%	-10.2%	-5.3%	1.0%	2.9%	10.2%
PCA-Squared (5 Components)	-55.0%	-42.1%	-32.5%	-22.6%	-16.4%	-5.5%
PCA-Squared (3 Components)	-46.8%	-38.8%	-30.6%	-21.0%	-17.0%	-8.9%
Partial Least Squares (5 components)	-56.3%	-40.5%	-33.4%	-25.7%	-18.6%	-9.5%
Partial Least Squares (3 components)	-57.9%	-40.7%	-31.9%	-22.9%	-14.6%	-1.8%
Panel B: Penalized Linear Regressions						
Ridge	-40.1%	-25.8%	-18.9%	-11.4%	-6.0%	4.6%
Lasso	-11.5%	-8.1%	-2.5%	0.2%	2.3%	9.3%
Elastic Net	-10.7%	-8.4%	-5.1%	0.4%	0.3%	7.0%



PCA, PLS, Penalised Linear Regressions, Bond return predictability and macroeconomic variables

What can be seen:

The performance of sparse linear models stands out. Unlike the yield-only case, Elastic Net delivers positive R^2 across maturities, especially when forward rates enter as a single CP factor rather than ten separate regressors. Dense methods, PCR with eight macro PCs and PLS continue to perform poorly out-of-sample.

Interesting, the way forward rates enter the regression matters. Restricting them to a single Cochrane–Piazzesi factor improves Elastic Net's out-of-sample accuracy, showing that macro information adds predictive power beyond the yield curve.

Models	R^2_{OOS}					
	$xr_{t+1}^{(2)}$	$xr_{t+1}^{(3)}$	$xr_{t+1}^{(4)}$	$xr_{t+1}^{(5)}$	$xr_{t+1}^{(7)}$	$xr_{t+1}^{(10)}$
Panel A: PCA and PLS						
PCA - first 8 PC'S	-9.8%	-2.9%	0.3%	3.0%	3.3%	4.5%
PCA as in Ludvigson and Ng (2009)	-3.4%	0.2%	1.6%	1.6%	-1.4%	-4.7%
PLS - 8 components	-40.7%	-19.7%	-12.0%	-8.2%	-2.7%	3.4%
Panel B: Penalized Linear Regressions						
Ridge (using CP factor)	-45.3%	-23.6%	-16.7%	-13.2%	-3.1%	5.3%
Lasso (using CP factor)	6.4%	11.2%	12.9%	14.4%	19.6%	23.7%
Elastic Net (using CP factor)	6.4%	11.0%	14.3%	15.7%	21.7%	29.1%
Ridge (using fwd rates directly)	-52.2%	-28.7%	-22.7%	-18.3%	-13.1%	-3.5%
Lasso (using fwd rates directly)	11.0%	12.0%	12.3%	16.4%	19.9%	23.6%
Elastic Net (using fwd rates directly)	10.2%	14.2%	16.0%	13.2%	19.9%	23.6%



1. Forecast with forward rates (Table 1)

Модель	rx^2	rx^3	rx^4	rx^5	rx^7	rx^{10}	Портфель
PCA (10 components)	-51.2%	-36.9%	-28.8%	-22.9%	-16.0%	-5.2%	-21.0%
PCA (5 components)	-51.7%	-38.1%	-30.9%	-23.7%	-19.0%	-7.8%	-23.3%
PCA (3 components)	-19.1%	-12.4%	-8.1%	-2.9%	0.3%*	7.1%*	-2.2%
PCA-Squared (5 components)	-53.3%	-41.7%	-32.2%	-24.8%	-18.6%	-8.5%	-24.3%
PCA-Squared (3 components)	-48.0%	-39.4%	-31.1%	-24.1%	-18.3%	-10.0%	-23.9%
Partial Least Squares (5 components)	-51.8%	-36.8%	-29.1%	-22.9%	-14.3%	-3.0%	-19.8%
Partial Least Squares (3 components)	-52.0%	-38.5%	-31.3%	-24.6%	-18.1%	-6.4%	-22.7%
Ridge	-11.4%	-8.6%	-7.8%	-3.5%	-1.3%	2.3%*	-2.5%
Lasso	-9.3%	-6.8%	-3.5%	0.2%*	1.9%*	6.9%*	1.7%*
Elastic Net	-10.9%	-6.2%	-4.6%	-1.0%	2.1%*	3.3%*	0.3%*

Out-of-sample R^2 relative to the expanding historical mean. An asterisk denotes 10% Clark–West significance.



2. Forecast with forward rates and the macro panel (Table 2)

Модель	rx^2	rx^3	rx^4	rx^5	rx^7	rx^{10}	Портфель
PCA - first 8 PCs	-6.4%	-4.5%	-4.3%	-5.0%	-5.2%	-6.3%	-6.4%
PCA as in Ludvigson and Ng (2009)	0.3%*	0.8%*	1.6%*	0.3%*	-2.9%	-7.6%	-3.3%
PLS - 8 components	-98.7%	-84.4%	-100.1%	-173.3%	-180.2%	-260.8%	-179.8%
Ridge (using CP factor)	4.8%*	7.6%*	6.7%*	8.2%*	6.9%*	9.4%*	8.9%*
Lasso (using CP factor)	6.3%*	16.2%*	18.0%*	20.0%*	19.1%*	22.7%*	22.9%*
Elastic Net (using CP factor)	8.0%*	14.3%*	12.3%*	11.3%*	14.9%*	19.2%*	17.6%*
Ridge (using fwd rates directly)	-9.0%	-10.5%	-7.4%	-5.0%	-4.0%	5.5%*	-1.4%
Lasso (using fwd rates directly)	-8.7%	-7.1%	-4.4%	-3.4%	-1.6%	7.9%*	1.1%*
Elastic Net (using fwd rates directly)	-5.8%	-8.3%	-6.2%	-6.1%	-5.6%	4.3%*	-2.6%

Out-of-sample R^2 relative to the expanding historical mean. An asterisk denotes 10% Clark–West significance.

3. Tree-based models

All three models use regression trees to approximate nonlinear relations between predictors and excess bond returns.

GBRT: sequentially adds shallow trees, with each new tree correcting the remaining forecast errors.

Random Forest: averages fully grown bootstrap trees built on random subsets of predictors.

Extra Trees: averages fully grown trees with randomized split thresholds and no bootstrap sampling.

Parameter choices

The paper uses unrestricted-depth trees for Random Forest and Extra Trees, while GBRT relies on shallow trees.

The remaining settings were selected or fixed in our notebook using validation grids.

FIGURE 1: Example of a Regression Tree

This figure shows an example of a regression tree for a predictive regression with a univariate target variable, e.g., the holding period excess return of a one-year treasury bond, and two predictors, e.g., the two-year and the five-year forward rates, which we label $y_t^{(2)}$ and $y_t^{(5)}$. Left panel shows the partition of the two-dimensional regression space by recursive splitting. Right panel shows the corresponding regression tree.

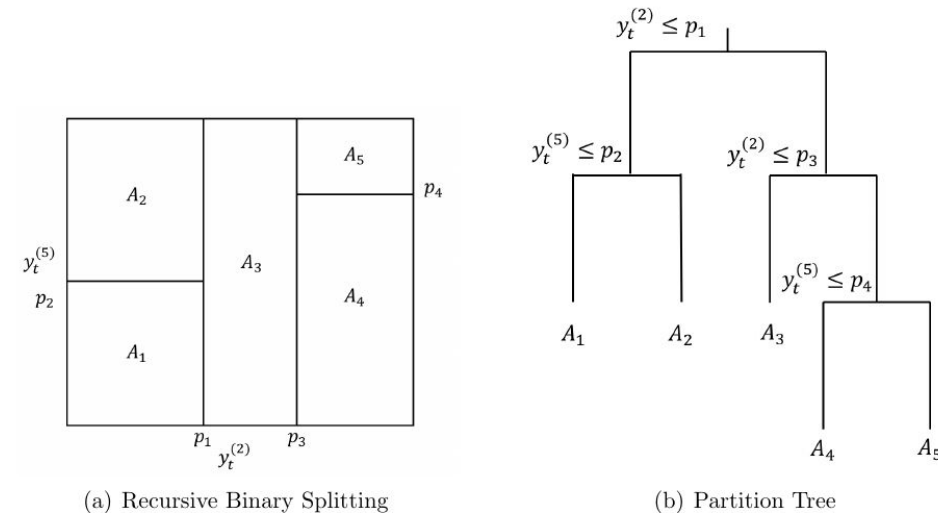


Figure 1. Recursive binary splitting and the corresponding regression tree



Bond return predictability: paper results

Forward rates only

Panel C: Regression Trees and Neural Networks

Gradient Boosted Tree	4.9%	4.9%	7.0%	10.8%	7.1%	11.2%
Random Forest	12.2%	13.1%	15.2%	17.4%	14.9%	16.0%
Extreme Tree	3.9%	10.1%	14.3%	16.9%	19.7%	24.6%

Forward rates and macroeconomic variables

Panel C: Regression Trees and Neural Networks

Gradient Boosted Tree	13.1%	15.9%	18.1%	23.8%	22.5%	25.5%
Random Forest	26.7%	21.5%	22.0%	24.4%	20.0%	25.0%
Extreme Tree	23.0%	23.4%	22.3%	23.7%	29.9%	29.6%

The paper reports positive R^2_{oos} for all tree methods. Adding macroeconomic variables raises predictive accuracy, especially for longer maturities.

Main findings:

- The qualitative pattern is similar across both specifications: tree methods detect positive and economically meaningful predictability.
- The article motivates the same comparison used in our notebook: yield-curve information alone versus yield-curve information augmented with macro variables.



Our results for Tree-Based Models

Forecasting with forward rates only

Recursive out-of-sample forecasts: January 1990 to December 2018. Hyperparameters were selected monthly using an 85/15 train-validation split.

Model	xr_2y	xr_3y	xr_4y	xr_5y	xr_7y	xr_10y
GBRT	+4.37	+6.72	+6.84	+10.44	+10.72	+16.54
Random Forest	+8.18	+11.86	+14.06	+16.73	+17.32	+21.14
Extra Trees	+5.43	+7.60	+9.38	+11.59	+13.98	+18.11

Main findings:

- The results are broadly close to the paper, with deviations of a few percentage points in most maturities.
- As in the paper, R^2_{oos} is generally higher for longer maturities, with occasional local declines rather than a perfectly monotonic path.
- All three models produce positive out-of-sample R^2_{oos} .



Our results for Tree-Based Models

Forecasting with forward rates and macroeconomic variables

The predictor set combines forward rates with transformed FRED-MD macroeconomic variables.

Model	xr_2y	xr_3y	xr_4y	xr_5y	xr_7y	xr_10y
GBRT	+13.62	+14.68	+16.11	+17.74	+16.64	+19.97
Random Forest	+28.79	+24.48	+23.66	+24.50	+25.86	+25.65
Extra Trees	+22.94	+24.07	+21.93	+27.60	+31.92	+34.03

Main findings:

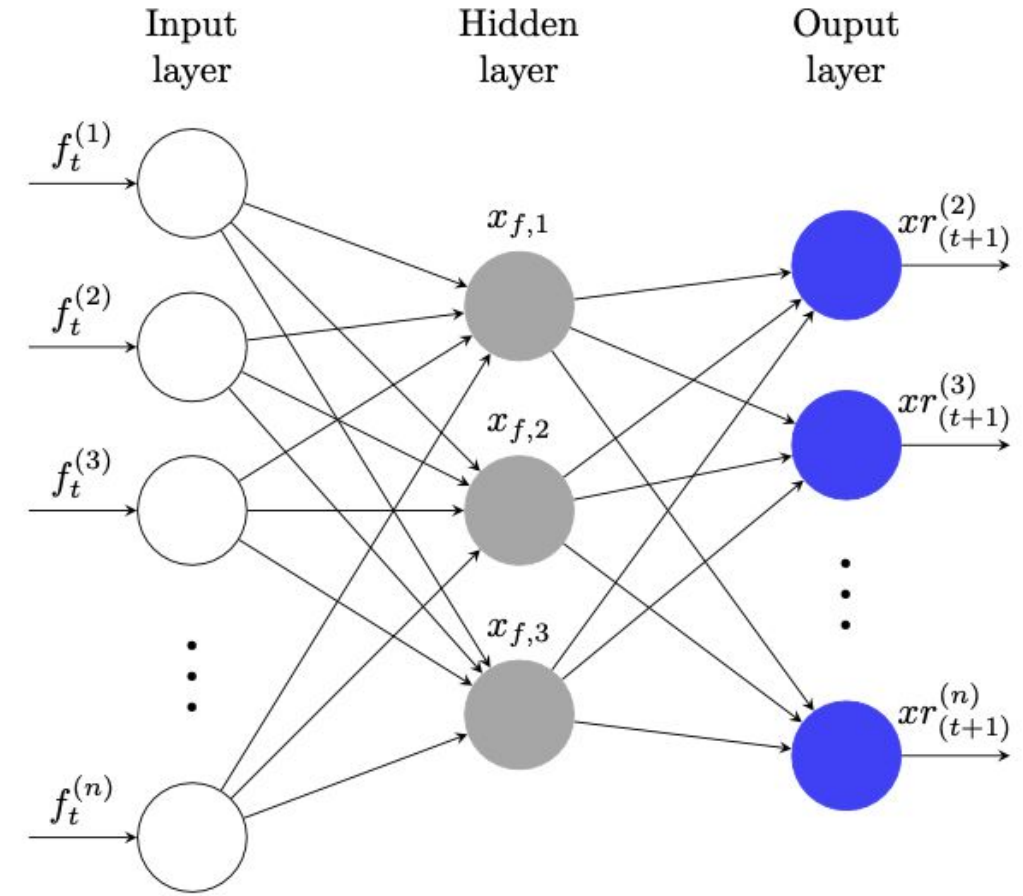
- Adding macroeconomic variables increases the level of R^2_{oos} for all three models, as in the paper.
- The maturity profiles broadly reproduce the paper: performance generally rises with maturity, with some local dips.
- The estimates are of a similar magnitude to the paper, although a few maturities show larger deviations due to our validation grids and implementation.

4. Neural networks

Neural networks: yields-only

To forecast bond returns using only information from the term structure of interest rates, use variants of architecture as depicted in Fig. 1.

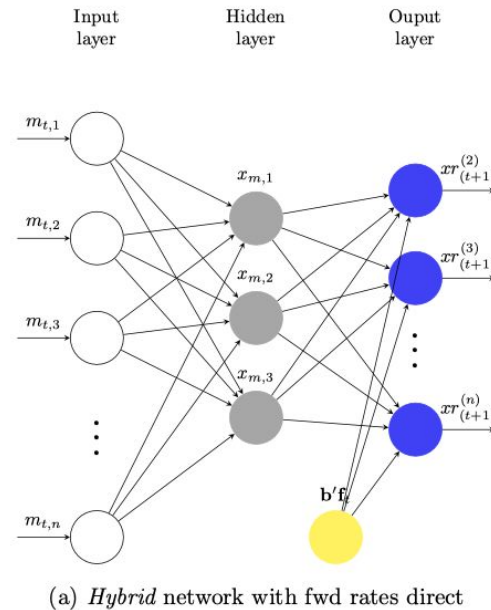
Consists of at least three layers of nodes, an input layer, a hidden layer, an output layer.



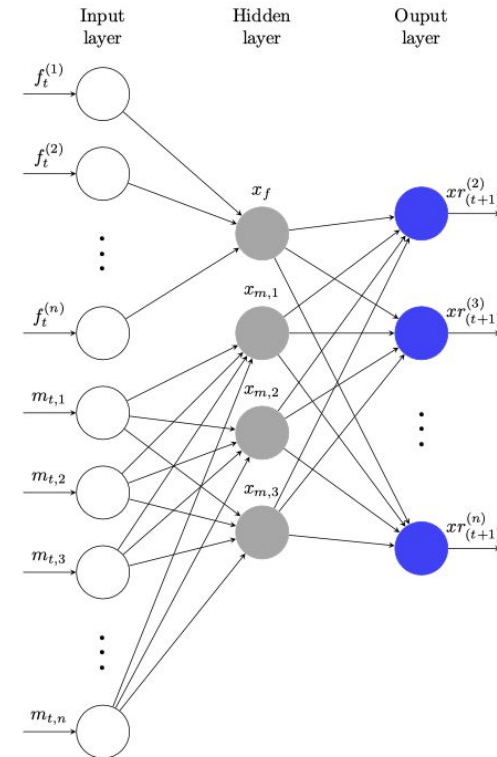
4. Neural networks

Neural networks: macro plus yields

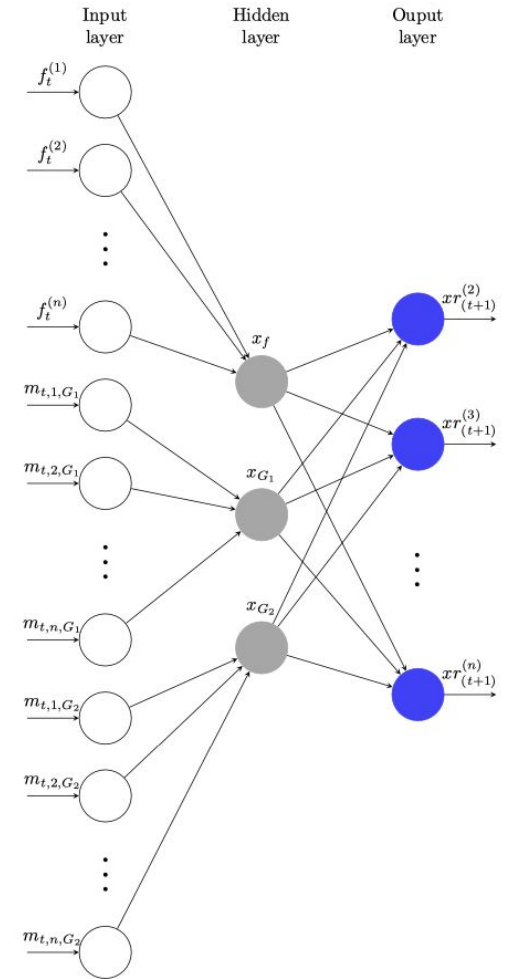
- “Hybrid”: forward rates are included as additional predictor in the output layer.
- Macro network + fwd rate net: ensembles two separate networks at the output layer level.
- Group-ensemble network: entails a collection of networks, one for each group of macroeconomic variables, which are trained in parallel and ensembled at the output layer level



(a) Hybrid network with fwd rates direct



(b) Macro network + fwd rate net



(c) Group-ensemble network



Neural Networks, Bond return predictability and the yield curve

What can be seen:

- NN 2 Layer (7 nodes each) outperforms others for all maturities
- NNs with 3 Layers show poor results
- NN with 1 Layer (3 nodes) perform on par with the best, deeper network
- Last row reports the results obtained by feeding NN with the ten forward rates at time t and lagged forward rates from time $t - 11$ to $t - 1$. By comparing to other results, there is no evidence that we can improve upon a NN that uses just the month- t forward rates

Models	R^2_{oos}					
	$xr_{t+1}^{(2)}$	$xr_{t+1}^{(3)}$	$xr_{t+1}^{(4)}$	$xr_{t+1}^{(5)}$	$xr_{t+1}^{(7)}$	$xr_{t+1}^{(10)}$
NN - 1 Layer (3 nodes)	12.7%	16.4%	19.5%	21.6%	23.1%	26.4%
NN - 1 Layer (5 nodes)	7.0%	10.7%	14.6%	17.2%	18.9%	22.9%
NN - 1 Layer (7 nodes)	-3.2%	3.0%	7.1%	12.1%	13.6%	17.1%
NN - 2 Layer (3 nodes each)	7.4%	10.8%	12.9%	14.7%	16.3%	19.3%
NN - 2 Layer (5 nodes each)	8.9%	13.0%	16.0%	18.5%	20.4%	23.6%
NN - 2 Layer (7 nodes each)	9.0%	15.2%	18.1%	21.2%	23.7%	27.4%
NN - 3 Layer (3 nodes each)	3.4%	8.9%	10.9%	11.2%	12.7%	15.1%
NN - 3 Layer (5 nodes each)	3.6%	7.4%	8.9%	10.6%	11.6%	13.7%
NN - 3 Layer (7 nodes each)	3.9%	6.7%	8.3%	8.9%	11.1%	13.2%
NN - 3 Layer (5,4,3 nodes each)	-0.9%	2.8%	5.6%	6.5%	8.3%	10.6%
NN - 1 Layer (7 nodes), lagged inputs: $t - 1 : t - 11$	6.4%	14.4%	17.0%	19.5%	20.3%	24.4%



Neural Networks, Bond return predictability and macroeconomic variables

What can be seen:

- The performance of hybrid networks stands out. Differently from Table 1, increasing the depth of the NN from one to three layers improves its accuracy.
- Still, careful choice of network structure based on prior economic information exerts a great impact on performance. One-layer group ensembles model performs on par with the three-layer hybrid NN for short and medium maturities and attains the highest predictive accuracy for the 7 and 10 year bonds.

Models	R^2_{bonds}					
	$xr_{t+1}^{(2)}$	$xr_{t+1}^{(3)}$	$xr_{t+1}^{(4)}$	$xr_{t+1}^{(5)}$	$xr_{t+1}^{(7)}$	$xr_{t+1}^{(10)}$
NN 1 Layer (32 nodes), fwd rates direct	6.0%	13.6%	17.8%	22.0%	22.5%	26.5%
NN 2 Layer (32, 16 nodes), fwd rates direct	16.6%	21.5%	24.9%	28.0%	29.4%	32.3%
NN 3 Layer (32, 16, 8 nodes), fwd rates direct	24.8%	26.3%	29.7%	32.0%	31.7%	33.7%
NN 1 Layer (32 nodes), fwd rates net (1 layer: 3 nodes)	8.4%	19.0%	23.8%	25.6%	27.2%	29.5%
NN 2 Layer (32,16, nodes), fwd rates net (1 layer: 3 nodes)	12.1%	15.7%	20.0%	23.5%	25.4%	28.1%
NN 3 Layer (32,16, 8 nodes), fwd rates net (1 layer: 3 nodes)	7.6%	16.3%	20.2%	23.7%	25.0%	28.1%
NN 1 Layer Group Ensem (1 node per group), fwd rates direct	12.6%	17.3%	21.6%	24.2%	25.9%	29.6%
NN 1 Layer Group Ensem (1 node per group), fwd rates net (1 layer: 3 nodes)	20.0%	25.6%	29.5%	31.2%	33.6%	36.3%
NN 2 Layer Group Ensem (2,1 nodes per group / hidden layer), fwd rates net (2 layer: 3 nodes)	17.3%	23.6%	27.8%	29.8%	31.0%	33.0%
NN 3 Layer Group Ensem (3, 2, 1 nodes per group / hidden layer), fwd rates net (3 layer: 3 nodes)	13.6%	20.0%	23.7%	26.1%	27.5%	30.7%



Our results for Neural Networks

Forecasting with forward rates:

- To reduce learning time, n_seeds were changed from 100 to 50, epochs from 1000 to 500

	xr_2y	xr_3y	xr_4y	xr_5y	xr_7y	xr_10y
NN 2 Layer (7 nodes each)	+2.62	+3.18	-4.91	+5.60	+7.43	+7.12
NN 1 Layer (7 nodes)	-16.04	+0.99	+2.18	+1.98	+1.63	+6.39
NN 1 Layer (7 nodes), lagged inputs 1..11	-22.17	-5.62	-3.91	-1.29	-0.99	+0.20

Main findings:

- NN 2 Layers (7 nodes) outperforms others, as should be
- Lagged inputs do not help
- R^2 increases with maturity for all architectures



Our results for Neural Networks

Forecasting with forward rates and macroeconomic variables:

- To reduce learning time, n_seeds were changed from 100 to 50, epochs from 1000 to 500

	xr_2y	xr_3y	xr_4y	xr_5y	xr_7y	xr_10y
NN 1 Layer Group Ensem + fwd rate net	+6.63	+8.91	+6.38	+4.49	+4.52	+7.17
NN 1 Layer (32 nodes), fwd rates direct	-47.15	-43.81	-82.20	-45.87	-50.92	-44.64

Main findings:

Macro variables alone do not help. How you structure the network is what matters.

- Group Ensemble with small sub-networks, one per economic category works
- 1 Layer (32) - one flat network over all 125 macro inputs breaks completely
- Careful choice of architecture for macroeconomic variables helps to outperform those with just forward rates (+8.91 > +3.18 for xr_3y)

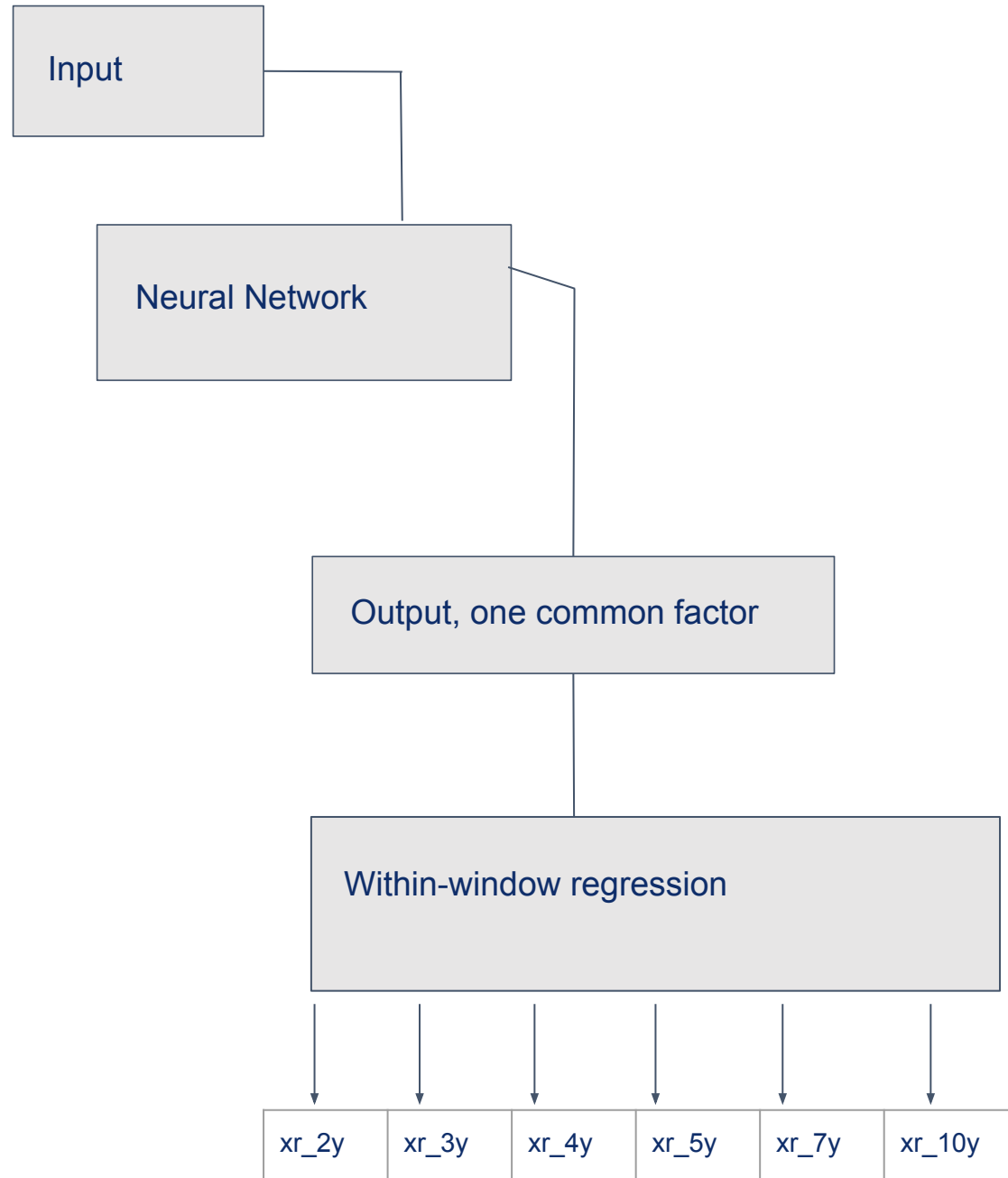


The Experiment: Single-Output Architecture

The six target maturities are almost perfectly described by one common factor: the first principal component explains 95.5% of their variance.

Instead of predicting six returns independently, the network now predicts one common factor. This factor is mapped back to each maturity through loadings estimated by a within-window regression: for each origin, we regress the realized excess returns on the network's factor using the training window, and use the resulting coefficients to translate the factor into maturity-specific forecasts.

This reform raised the portfolio R^2 from 3.8% to 13.9% and the ten-year R^2 from -0.5% to $+17.3\%$, with the Clark–West test becoming significant at every maturity. The improvement replicates under both target-scaling variants.





The Experiment: Single-Output Architecture

Results

- Base model: NN 1 Layer (3 nodes), six independent outputs.
- Reform: the same network, but with one output - a single common risk-premium factor.
- Maturity mapping: the factor is mapped back to each maturity through loadings from a within-window regression.
- Result: R^2 rises from +3.8% to +13.9% at the portfolio level, and from -0.5% to +17.3% at 10 years.

	rx^2	rx^3	rx^4	rx^5	rx^7	rx^{10}	EW
6 output nodes NN, adaptation	-38.6	-9.4	+0.4	+4.5	+3.0	-0.5	+3.8
One output	-0.1	+6.2	+10.0	+12.6	+14.7	+17.3	+13.9

Setup: 10 seeds with top-3 selection (vs. 100/top-10 in the paper). The improvement is expected to persist with more seeds too.



Limitations and Open Questions

Methodological choices

- CV of hyperparameters every 5 years is arbitrary.
- Seed policy is not fully specified - how the 100 seeds are drawn and what happens on poor initialization are not described.
- No explanations on why exact hyperparameters were chosen

Data

- Long-maturity yields are kernel-smoothed - the 20y and 30y points are interpolated, not observed.
- Campbell–Thompson R^2 is not protected against overfitting - no placebo test on shuffled targets.

Model and results

- The "deep learning" label is misleading - the model is a 1-layer, 3-node, maximum 3 layers.
- Depth does not help - no evidence that LSTM, GRU, or ResNet-style models were tested.
- SGD + Nesterov instead of Adam - unusual for 2020, may understate NN performance.

Reproducibility

- The code is not published — exact implementation, hyperparameter search, and model combination are not reproducible.



Conclusion

What the paper found

- ML methods - extreme trees and neural networks - predict bond risk premia; linear models mostly fail.
- Macro information is unspanned by the yield curve: adding 128 macro variables raises R^2 by ~10 pp at every maturity.
- Structure beats depth: a 1-layer group-ensembled NN matches a 3-layer hybrid network.
- Lags do not help shallow networks.

What we replicated

- Replicated all main algorithms and got similar or close result

Our improvement

- The six targets are almost a single factor.
- Replaced 6 outputs with 1 common factor mapped back through within-window loadings.
- EW R^2 rose from +3.8% to +13.9%, 10y from -0.5% to +17.3%.

